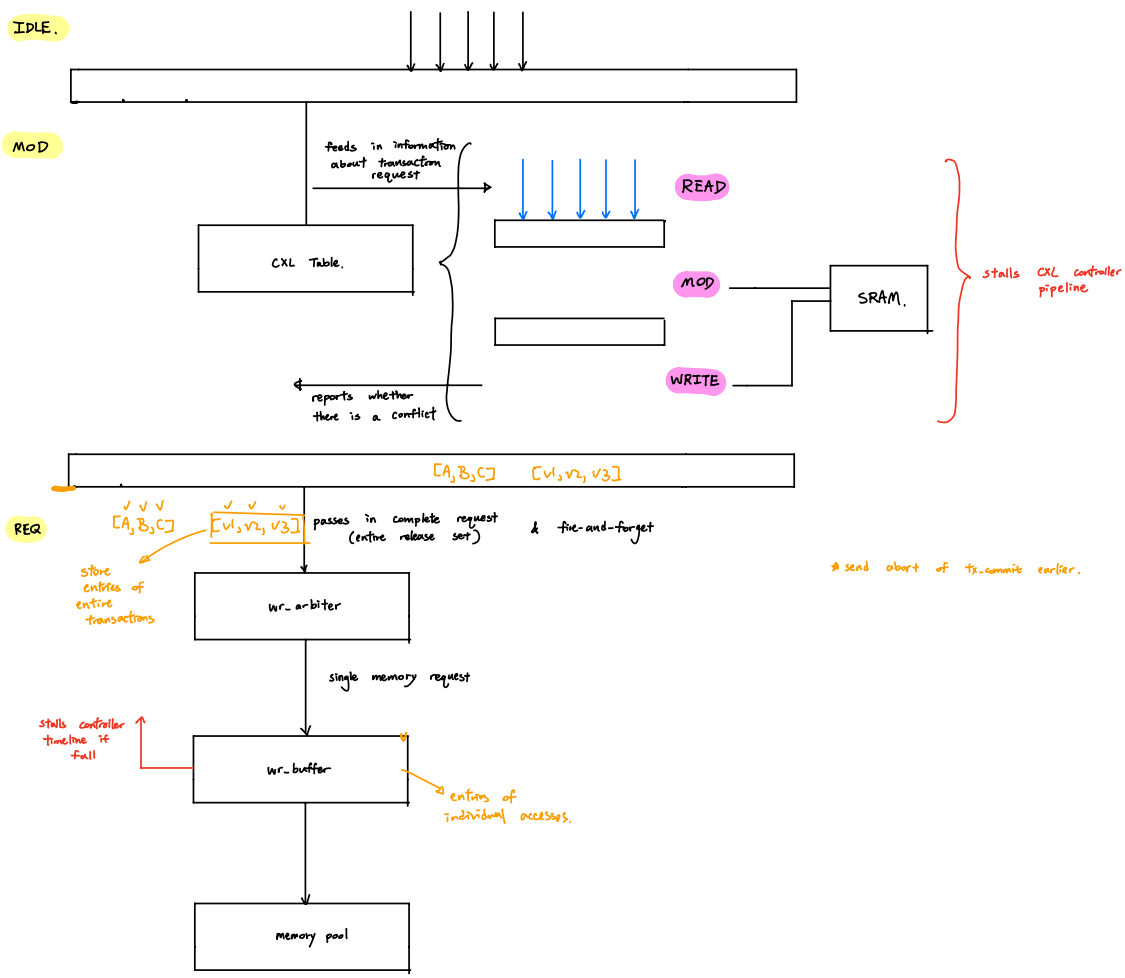


Examining stall times from output arbiter:

=> A LOAD takes 7 clock cycles to exit pipeline

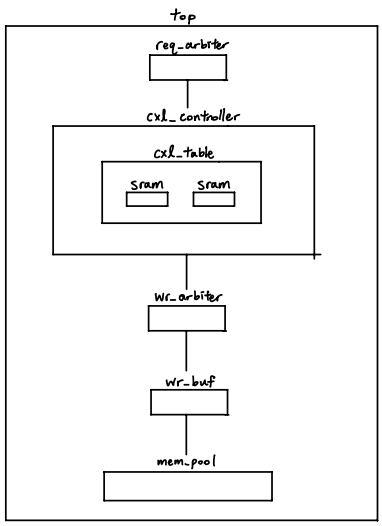
=> A COMMIT with 16 full entries takes 22 clock cycles to exit pipeline

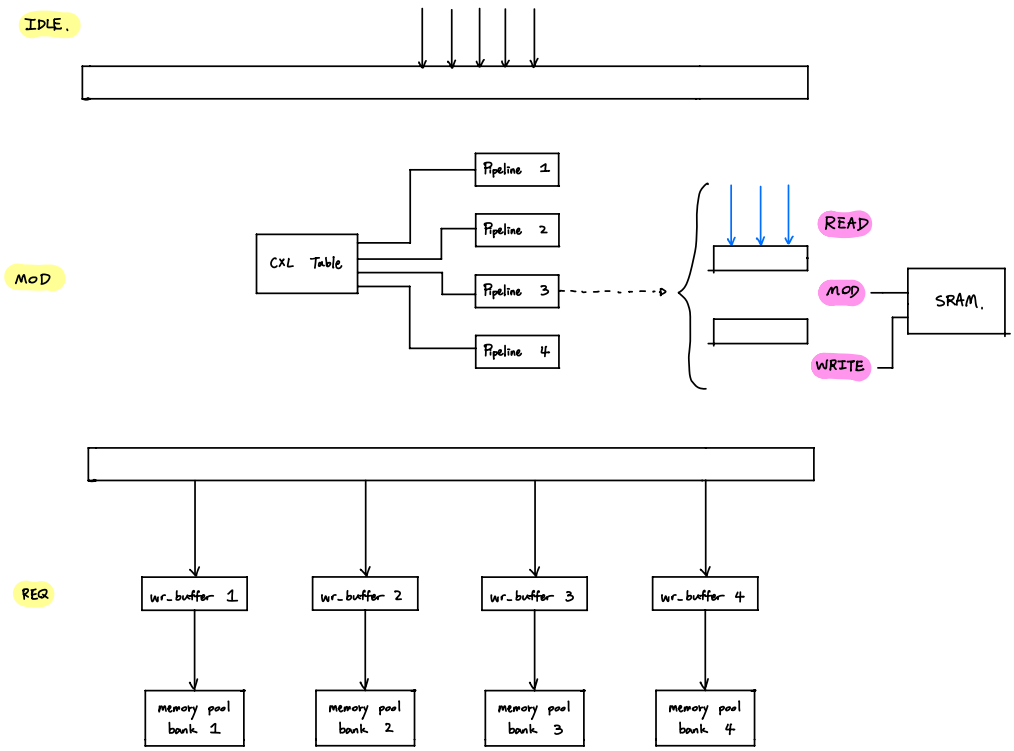
① LOAD , LOAD → no bottleneck

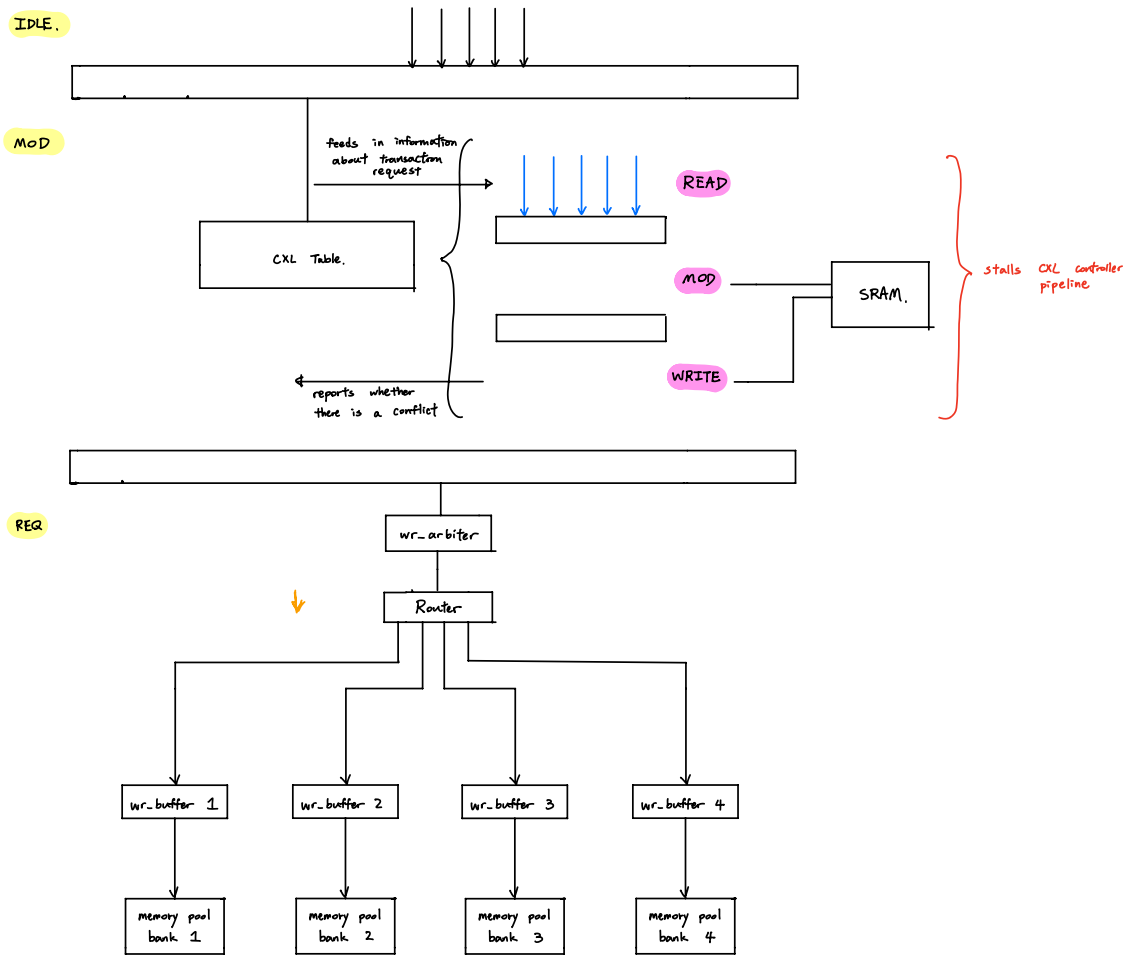
② LOAD , COMMIT → no bottleneck

③ COMMIT , COMMIT → bottleneck (depending on COMMIT size)

④ COMMIT, LOAD → bottleneck (depending on COMMIT size)


top
req_arbiter
cxl_controller
cxl_table
sram
sram
wr_arbiter
wr_buf
mem_pool


IDLE.
MOD
Pipeline 1
Pipeline 2
Pipeline 3
Pipeline 4
CXL Table
READ
MOD
WRITE
SRAM.
REQ
wr_buffer 1
wr_buffer 2
wr_buffer 3
wr_buffer 4
memory pool bank 1
memory pool bank 2
memory pool bank 3
memory pool bank 4


IDLE.
MOD
feeds in information about transaction request
READ
CXL Table.
MOD
SRAM.
stalls CXL controller pipeline
WRITE
reports whether there is a conflict
REQ
wr_arbiter
Router
wr_buffer 1
wr_buffer 2
wr_buffer 3
wr_buffer 4
memory pool bank 1
memory pool bank 2
memory pool bank 3
memory pool bank 4